Evaluation Graphs and Explanations

IEEE-style figures generated from recorded transcription and consistency-validation data

Purpose and evidence boundary

This document contains publication-ready graphs for the Meeting Analyzer evaluation. The figures use the measured transcription benchmark in research_evaluation/data and the recorded five-run consistency audit described in REANALYSIS_CONSISTENCY_REPORT.md. Values are not inferred from the architecture diagram and no unsupported accuracy values are added.

The transcription benchmark contains one source recording, three clip durations (30, 60, and 120 seconds), and three repeated runs per duration. The consistency fixture contains five fresh analyses of the same meeting. These are engineering validation results, not a corpus-wide accuracy study.

Tables for the IEEE manuscript

The following tables report configuration and measured results directly. Jaccard overlap has been removed from this PDF.

TABLE I. SYSTEM TECHNOLOGY STACK

Layer	Technology	Role
Frontend	React, Vite, Tailwind CSS	Interactive dashboard
Backend	FastAPI, Python	API and orchestration
Audio	FFmpeg	Extraction and normalization
Speech recognition	Faster-Whisper	Timestamped transcription
NLP	spaCy, NLTK	Preprocessing and linguistic analysis
Classification	TF-IDF + Logistic Regression	Dialogue-act classification
Topics	LDA / NMF	Topic discovery
Storage	Local JSON / Supabase	Persistence and recovery
PDF export	ReportLab	Report generation

TABLE II. DATASET AND EVALUATION CONFIGURATION

Parameter	Value
Source recordings	1
Clip durations	30 s, 60 s, 120 s
Transcription repeats	3 per duration
Re-analysis runs	5
Whisper model	Systran/faster-whisper-base
Device / compute	CPU / int8
Beam size	5
VAD	Enabled; 500 ms minimum silence
Language	Automatic detection

TABLE III. TRANSCRIPTION LATENCY UNDER CPU/INT8 EXECUTION

Duration (s)	Mean (s)	SD (s)	Minimum (s)	Maximum (s)	Mean RTF
30	2.315	0.483	1.757	2.614	0.077
60	5.715	0.563	5.065	6.041	0.095
120	11.562	0.537	11.063	12.130	0.096

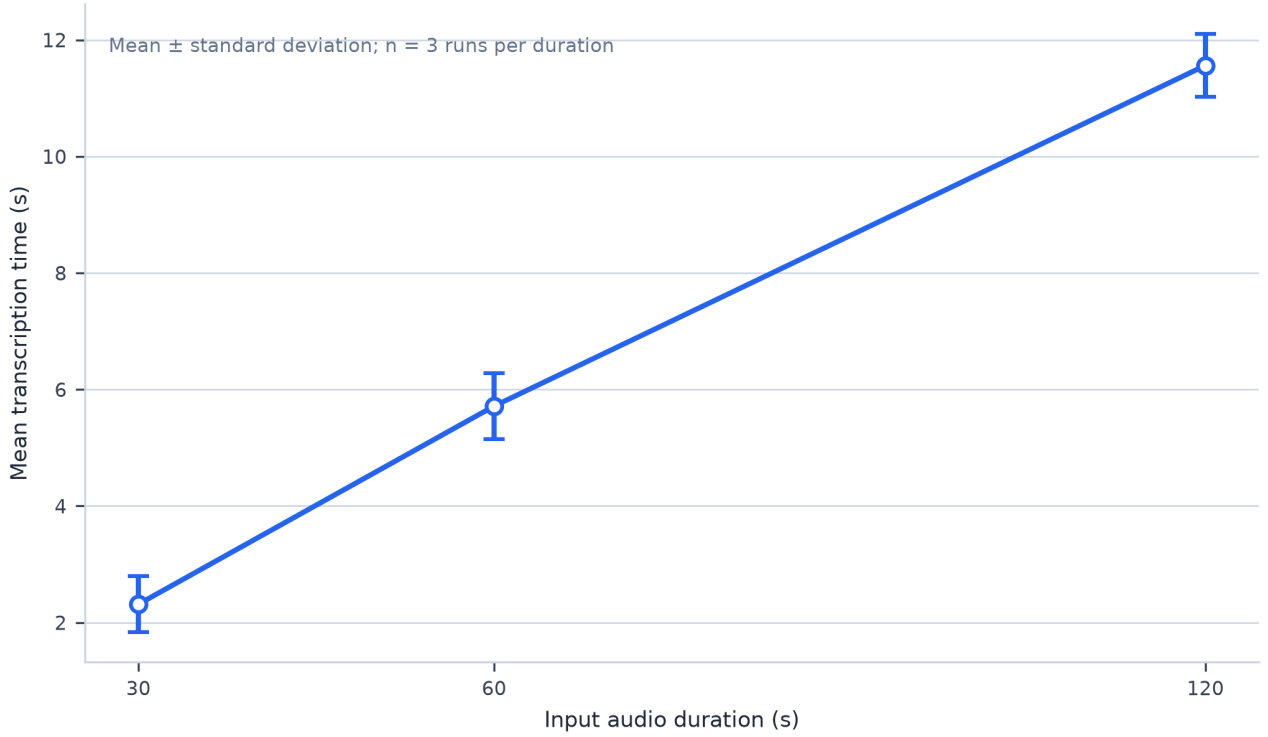


Fig. 1. Mean transcription latency with one-standard-deviation error bars. Latency increases approximately linearly with clip duration for the measured CPU/int8 Faster-Whisper benchmark.

Interpretation. The graph reports wall-clock time for warm transcription, including decode, VAD, language detection, segment consumption, and output construction. Model loading, NLP, persistence, and PDF generation are excluded.

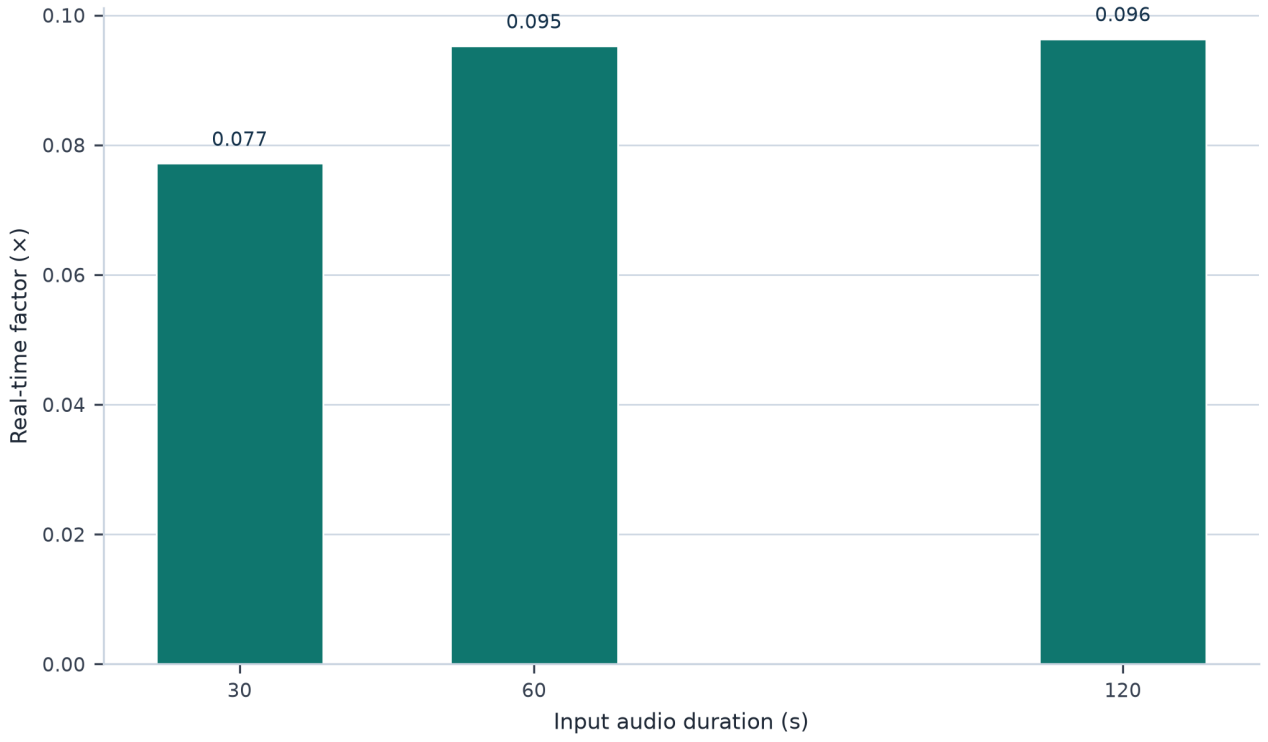


Fig. 2. Speech-to-text real-time factor for the three measured input durations. Values below 1.0 indicate processing faster than the duration of the audio.

Interpretation. The measured mean real-time factor remains near 0.08–0.10×, indicating that the tested CPU configuration processed each clip substantially faster than real time. This result is specific to the tested hardware and model settings.

Results reported as tables

The repeated extraction counts and validation-stage counts are constant for this fixture. Plotting constant series would add visual weight without adding information, so they are reported as compact tables instead.

TABLE IV. CORE EXTRACTION COUNTS ACROSS FIVE FRESH ANALYSES

Run	Actions	Decisions	Questions
1	2	1	1
2	2	1	1
3	2	1	1
4	2	1	1
5	2	1	1

The recorded Runs 1–2 retention audit also reported 2 actions, 1 decision, and 1 question at each of Generated, Parsed, Validated, Evidence matched, Deduplicated, and Persisted stages. This supports the retention claim without representing a single fixture as a general production guarantee.

TABLE V. VALIDATION PIPELINE RETENTION

Category	Generated	Parsed	Validated	Evidence matched	Deduplicated	Persisted
Action items	2	2	2	2	2	2
Decisions	1	1	1	1	1	1
Questions	1	1	1	1	1	1
Follow-ups	0	0	0	0	0	0

TABLE VI. SUMMARIZATION METHODS

Method	Algorithm / model	Output	Grounding
Extractive	TF-IDF + centroid similarity + MMR	Source excerpts	Direct transcript text
Abstractive	BART/T5-compatible Transformer	Generated draft	Requires human review
Semantic	Structured evidence extraction	Evidence-linked bullets	Verbatim evidence required

TABLE VII. LIMITATIONS AND SAFEGUARDS

Risk	Safeguard	Remaining limitation
Unsupported insight	Verbatim evidence matching	Lexical matching is not full entailment
Re-analysis drift	Versioned manifest and stable IDs	Fixture-level validation only
Speaker identity	Preserve available labels	No diarization for unlabeled audio
Long transcripts	Bounded chunks and deterministic merge	Higher resource requirements
Abstractive hallucination	Extractive default and review	Generated drafts need human review

Data provenance and reproducibility

The latency and real-time-factor figures are generated directly from research_evaluation/data/transcription_summary.csv. The source benchmark records the Faster-Whisper base model, CPU execution, int8 compute, beam size 5, VAD enabled, and three repeated runs per duration. The consistency figures use the documented recorded values: five stable core extraction runs and the stored action/decision/question overlap results.

Figure	Primary source	Sample
Figs. 1–2	research_evaluation/data/transcription_summary.csv	3 durations × 3 runs
Stability table	REANALYSIS_CONSISTENCY_REPORT.md	5 fresh analyses
Tables IV–V	REANALYSIS_CONSISTENCY_REPORT.md	5 runs; Runs 1–2 audit
Retention table	REANALYSIS_CONSISTENCY_REPORT.md	Recorded Runs 1–2 audit

Recommended paper placement: use Figs. 1–2 in the efficiency subsection and Tables IV–VII in the evaluation and discussion subsections, and retain the full explanations in an evaluation appendix if page limits are strict.